

ISHAAN SHARMA

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Education

Indian Institute of Technology Kharagpur

Nov 2022 – May 2026

B.Tech in Aerospace Engineering (Minor in Physics, Microspecialization in HPC)

CGPA 8.88/10.00 – **Class Rank 1**

Sharda Vidya Mandir Senior Secondary School, Bhopal

Session 2021-22

CBSE 12th Board Examination (AISSCE)

Aggregate 97%

Research Interests

My primary research interests lie in the study of compact objects, including binary interactions, merger dynamics, accretion flows, and multi-messenger astronomy. I am also enthusiastic about numerical relativity and computational astrophysics more broadly, including its applications to observational astronomy and cosmology.

Manuscripts in Preparation

Sharma, I., Kar, S., “Dynamical Stability and Tidal Disruption Conditions in Black Hole-Neutron Star Binaries via the Relativistic Affine Model”

A comprehensive stability analysis of neutron stars in BHNS binaries within the relativistic affine framework, incorporating realistic equations of state to derive tidal disruption criteria and characterize eigenmode structure.

Sharma, I., “Quadrupolar Spectral Signatures of Spiral Shock Instabilities in Relativistic Accretion Flows”

Time-dependent GRHD simulations of non-axisymmetric accretion flows in Schwarzschild spacetime, investigating nonlinear spiral shock dynamics, harmonic mode coupling, and quadrupolar spectral signatures via relativistic quadrupole extraction.

Research Projects

Tidal Effects in Black Hole–Neutron Star Mergers: A Simplified Model

Aug 2025 – May 2026

Bachelor’s Thesis | Supervisor: Prof. Sayan Kar, IIT Kharagpur

- Constructed quasiequilibrium sequences for close BH–NS binaries using the Relativistic Affine Model (RAM), characterizing the hydrodynamic equilibrium of ellipsoidal stars with TOV potential in Schwarzschild spacetime.
- Performed linear stability analysis and studied stellar **eigenmode evolution** for APR, SLy, and DD2 equations of state, investigating the relationship between eigenmode softening, dynamical stability, and neutron-star tidal disruption.
- Computed **tidal disruption radii** and post-merger remnant disk mass fractions across mass-ratio and compactness parameter space using fluid energy and four-velocity criteria at the onset of disruption.

Spectral GW Signatures from Spiral Shock Instabilities in Accretion Flows

May 2025 – Present

Independent Project

- Simulated low-angular-momentum shocked Schwarzschild BH accretion flows using Athena++ (2D equatorial GRHD, HRSC, Kerr–Schild coordinates) across sub-Keplerian angular momenta on logarithmic grids up to 768×384 .
- Triggered and analyzed non-axisymmetric spiral standing shock instabilities via Fourier decomposition, tracking nonlinear growth of $m = 1$ **spiral modes**, harmonic generation, saturation, and mode-coupling-driven $m = 2$ structure.
- Built a Python pipeline using NumPy, SciPy, h5py, and FFT-based spectral differentiation to extract quadrupole spectra, compute **quadrupolar GW signatures**, and correlate frequencies with spiral shock rotation dynamics.

Nonlinear Evolution of Primordial Non-Gaussianities in the Matter Bispectrum

Mar 2025 – May 2025

Supervisor: Prof. Somnath Bharadwaj, IIT Kharagpur

- Developed a custom parallelized Particle-Mesh N-body solver using pyFFTW and Numba to simulate nonlinear gravitational clustering, structure formation, and Fourier-space mode coupling in matter distributions.
- Generated Gaussian and local-type primordial non-Gaussian initial conditions using CLASS transfer functions, **2LPT** displacement fields, and quadratic perturbations parameterized by f_{NL} to study signatures in large-scale structure.
- Computed the reduced matter bispectrum across equilateral, squeezed, and folded triangular configurations using shell-filtered Fourier estimators, quantifying survival of primordial non-Gaussian signatures against nonlinear coupling.

Implementation of CIPSI in PySCF for Electron Correlation

Mar 2026 – Apr 2026

Supervisor: Prof. Sabyasachi Mishra, IIT Kharagpur

- Motivated by parallels between many-body electron correlation and dense matter in neutron star interiors, implemented the CIPSI algorithm in PySCF via iterative subspace expansion with second-order Epstein–Nesbet perturbation weights, enabling adaptive determinant selection and **PT2 energy corrections**.
- Characterized the onset of strong static correlation near the dissociation limit by tracking the leading determinant weight and natural orbital occupation; benchmarked PT2-corrected energies against FCI across bond dissociation curves.

Robust Aerodynamic Shape Optimization

May 2024 – Jul 2024

Supervisor: Prof. Aravind Balan, IISc Bengaluru

- Demonstrated a variance-penalized robust design methodology for ensuring off-design performance, validating a generalized polynomial chaos (gPC) uncertainty framework within an in-house HDG CFD solver.
- Modeled parametric and turbulence-modeling uncertainties via gPC, computing statistical moments of lift and drag without Monte Carlo sampling; quantified the mean-performance vs. robustness trade-off on canonical airfoils.

Design of a Gel Fuel Injector

Aug 2024 – Nov 2024

Lab Project | Supervisor: Prof. Srinibas Karmakar, IIT Kharagpur

- Demonstrated a **threefold reduction** in average droplet diameter vs. a conventional injector via DPM-based ANSYS Fluent simulations for HPMC gel, validating a novel injector design exploiting the shear-thinning property of gel fuels.
- Designed the injector to mitigate ligament formation and blockage challenges common in gel fuel atomization, informed by a comprehensive survey of gel fuel rheology and atomization techniques.

Term Papers

Tunneling Model for Hawking Radiation

Oct 2025 – Nov 2025

- Reproduced Hawking temperature and tunneling rate derivations in the Parikh–Wilczek framework using Painlevé–Gullstrand coordinates, and extended the analysis to Kerr and Kerr–Newman black holes.

McVittie Spacetime: Black Holes in an Expanding Universe

Mar 2025 – Apr 2025

- Analyzed the causal structure, geodesics, and physical interpretation of the McVittie metric as an exact GR solution for a central mass in an expanding FLRW background, and applied the formalism to the Plummer sphere model.

Technical Skills

Scientific & Simulation Tools: Athena++, AthenaK, CLASS, nbbodykit, pyFFTW, PySCF, ANSYS Fluent, Mathematica, SageMath, MATLAB, Astropy

Programming Languages: C, C++, Fortran, Python, LaTeX, Java, HTML/CSS, SQL

High-Performance Computing: MPI, OpenMP, CUDA

Advanced Coursework

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|----------------------------------|------------------------------|------------------------------|
| • General Theory of Relativity | • Astronomy and Astrophysics | • Numerical Methods |
| • Advanced Topics in Gravitation | • Particle Physics | • High-Performance Computing |
| • Physical Cosmology | • Statistical Physics | • Fluid Mechanics |

Academic Achievements

Institute Silver Medal • IIT Kharagpur (expected) Jul 2026
Awarded for obtaining the **highest CGPA** in the undergraduate program in Aerospace Engineering.

GRE Physics 2025 • ETS Nov 2025
Scored **970/990 (92nd percentile)** – 100% in Electromagnetism, 94% in QM, 93% in CM.

JC Ghosh Memorial Endowment Award • IIT Kharagpur Jul 2025
Awarded for obtaining the **highest CGPA** in Aerospace Engineering after the 6th semester.

GATE Physics 2025 • IIT Roorkee Mar 2025
Secured **All India Rank 223**, placing in the **top 1%** nationwide.

IOQA 2022 • HBCSE-TIFR May 2022
Received Certificate of Merit as the **State Topper of Madhya Pradesh** in the Astronomy Olympiad.

KVPY Fellow • IISc Bengaluru Mar 2021
Selected for the prestigious KVPY Fellowship funded by the Government of India.

IOQM 2021 • HBCSE-TIFR Feb 2021
Qualified for the Indian National Mathematical Olympiad and received the **Certificate of Merit**.

NTSE Scholar • NCERT Jul 2021
Awarded the National Talent Search Scholarship by NCERT, Government of India.

INJSO 2020 • HBCSE-TIFR Mar 2020
Ranked among the **top 40 students in India**, qualifying for the OCSC selection camp.